Introduction to Phishing

Phishing is defined as a type of cyberattack implemented via social engineering scam where malicious parties trick people into giving away private data such as card details or personally identifiable information (PII) or into installing harmful software. This is often done through the guise of pretending to be a trusted source. “Phishing attacks remain the most prevalent and disruptive type of breach or attack (experienced by 85% of businesses and 86% of charities).” (Rizvi, 2026) is what Rizvi said in the governments annual 2025 cybersecurity review. This shows the current prevalence of phishing attacks in the UK and why it is necessary that awareness and the correct precautions need to be actioned.

Why is phishing so effective?

Phishing’s effectiveness is mainly due to two factors that can work independently but are most effective when paired together. The first of these 2 factors is fraudulently pretending to be a trusted party. The most common way this is done during a phishing attack is manipulating the domain/email, this is the act of manipulating a URL or an outbound email address to resemble a trusted site such as nnicrosoft.com being used to mimic microsoft.com. Simple plays on the way a URL is typed can make an illusion that the site that is asking for your information is legitimate, with some scam sites mimicking the format of the original trusted website. The second of these 2 factors is playing on the psychological factors of victims. As stated in Frontiers in Computer Science, “when people are working under stress, they tend to make decisions without thinking of the possible consequences and options”. (Alkhalil, Z., Hewage, C., Nawaf, L., & Khan, I. (2021). Phishing attacks: A recent comprehensive study and a new anatomy. Frontiers in Computer Science, 3, 563060. Doi: HTTPs://doi.org/10.3389/fcomp.2021.563060). Cyber attackers will take advantage of this sort of tactic, often through the guise of creating a false sense of urgency or threat, which can be presented through forms such as faking an account closure based on the emails attached subscribed services, pretending to be a banking agency and asking for account details to resolve an issue, or a delivery company claiming a fraudulent charge and asking for bank details to process a refund. This sort of stress creates an immediate sense of urgency and panic with unaware individuals which can cause them to not question the scam or take the time to make sure that this situation is a scam, such as missing a small misspelling in an email address sending the scam. Some attacks known as spear-phishing will use a more specialised attack to target specific users which can often have information used in the scam that can cause the individual to be more concerned. This artificial stress created by these scams make them so dangerous as the weakest point of any cyber security solution is always the end users, as even when measures are put in

place, human emotions can be manipulated by malicious attackers a lot easier than a robust system can be breached through other means.

How to recognise a phishing attack

With the advent of AI, phishing attacks can become more convincing to the end user by using generative AI to mimic company websites, email formats and methods of reaching end users. Certain “fully trained SE models can theoretically learn from each attack and gradually improve their success rate.” ("Digital deception: generative artificial intelligence in social engineering and phishing", 2024, The Artificial Intelligence Review, vol. 57, no. 12, pp. 324.). Which is a cause for concern when thinking about how phishing will affect internet use in the future but is also the reason end users should be made aware of the most common ways to spot these types of attacks as at least one of these indicators should be present in each of these attacks. The first of these is to check the format of the email, some scammers will try to replicate the format of official emails, comparing these to official emails and spotting for differences alongside spelling mistakes could be an instant indicator of a phishing attack. Since the overall knowledge of attackers is limited, they will often try to make a sense of urgency using places where a user is signed up to. Such as if a delivery company tries to call about a false order, check your account to see if this delivery is in fact there or if a bank contacts you, place separate call to the bank to make sure that this charge is indeed there. The final key indicator is a mistake in the URL or email. In a URL it is often good to check for the lock next to it indicating a secure website as attackers will often purchase cheap domains without any form of encryption as the information that they steal is often not needing to be encrypted as they are not bound to keeping your information secure. Emails don't have any form of https that is visible but people within a business with a good email security system will often have an indicator in the email telling them that this email is not from a trusted source and to not reply unless certain. However, the key identifier within URLs and emails that is unanimous in both is, as mentioned before, subtle misspellings or changes in the URL/email to make the site/email look like the host that it is mimicking. Using these 4 signs, users should be able to identify most phishing attacks sent to them and the best piece of advice when possibly encountering a phishing attack is to step back and look at whether the attack is: sent by a trusted source, the source should be collecting the details, and what the original site says is the case.

What should you do if you suspect you're a victim of phishing?

If an end user were to suspect that they may have been a target of phishing it is important to first of all stop all contact with the attacker, keeping an attacker in the dark as much as possible when trying to resolve the issue should be the first line of defence against this sort of attack. This is the same for both an individual and in an organisation with the one key difference being that end users in an organisation should also report this to the relevant IT team. After the communication with the attacker is severed, actions should be undertaken to make sure that any damage from this attack is minimised as much as possible. In the latest 2025 phishing attack trends report, APWG (Phishing activity trends reports (2025) APWG. Available at: https://apwg.org/trendsreports (Accessed: 16 October 2026).) states that 30.9% of all phishing attacks are attacks against online payments and banking with wire transfer attacks increasing by 33% compared to previous quarters. The high rate of which bank details are transferred means that, when you have suspected that a phishing attack has taken place, you should ask whether the attack was targeting any banking information or has asked for any form of money to be sent to the attacker, as this can be reversed with a phone call to the relevant bank/building society.

Reference List

1) Rizvi, S. and Fordham, E. (2026) *Cyber security breaches survey 2025*, *GOV.UK*. Available at: https://www.gov.uk/government/statistics/cyber-security-breaches-survey-2025/cyber-security-breaches-survey-2025 (Accessed: 16 October 2026).
2) Alkhalil, Z., Hewage, C., Nawaf, L., & Khan, I. (2021). Phishing attacks: A recent comprehensive study and a new anatomy. Frontiers in Computer Science, 3, 563060. doi:https://doi.org/10.3389/fcomp.2021.563060
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4) Phishing activity trends reports (2025) APWG. Available at: https://apwg.org/trendsreports (Accessed: 16 October 2026).